



This illustration from Guy de Chauliac's *Chirurgia Magna* shows patients with a variety of injuries, including a broken arm and an eye wound, visiting a surgeon.

“ A **SURGEON** WHO DOES NOT KNOW HIS **ANATOMY** IS **LIKE A BLIND MAN** CARVING A LOG. ”

Guy de Chauliac, French physician, in *Great Surgery*, 1363

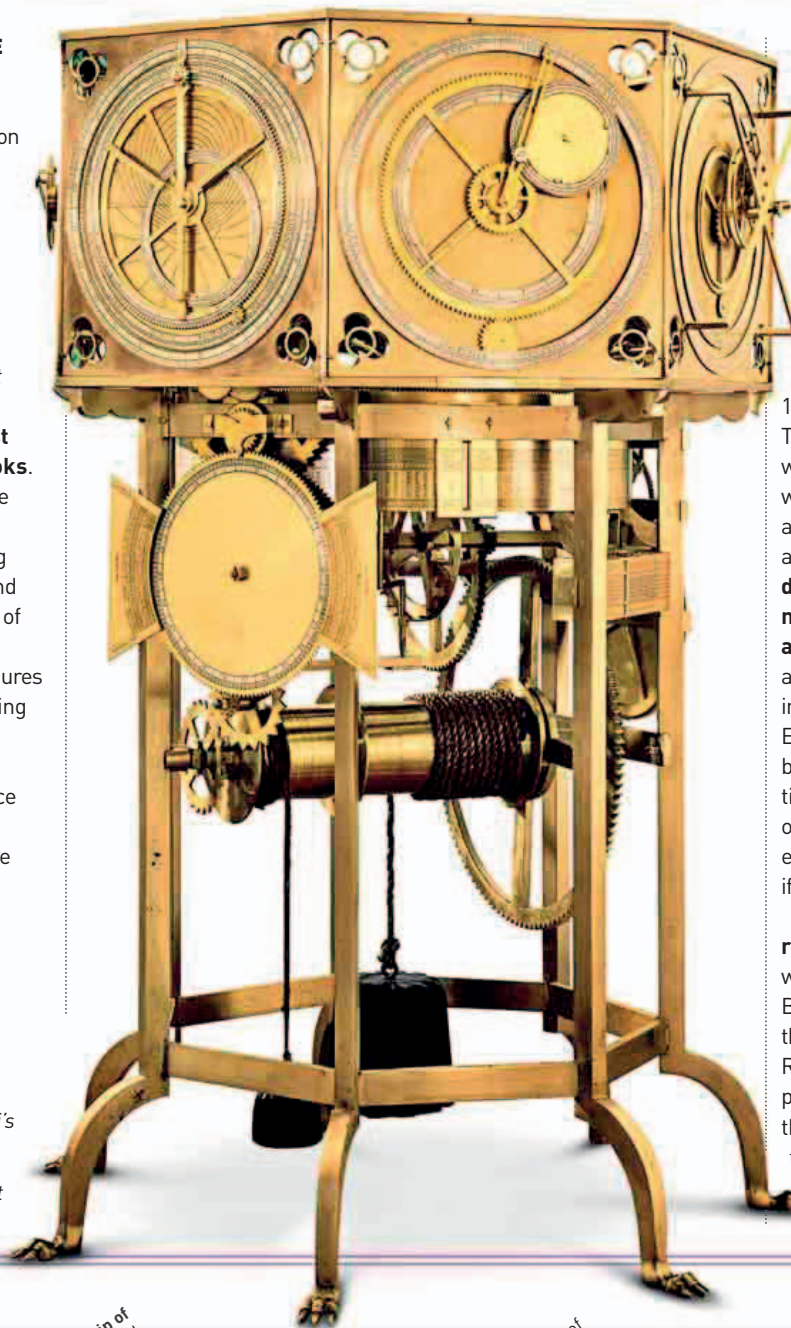
FRENCH PHYSICIAN GUY DE CHAULIAC

(c.1300–68) was personal surgeon to three Popes. He remained in Avignon during the outbreak of the Black Death in 1348, an experience that led him to **distinguish for the first time between pneumonic** (affecting the lungs) and **bubonic plague**.

His *Chirurgia Magna* (*Great Surgery*, 1363), became one of the medieval world's **most important surgical textbooks**. In its seven volumes, he gave advice on the treatment of fractures, advising extending broken limbs with pulleys and weights, and noting the loss of cerebrospinal fluid in skull fractures. He outlined procedures such as tracheotomies (cutting open the windpipe) and the replacement of lost teeth by ox bone. But his over-reliance on the work of Galen (see 100–250 BCE) led him to some retrograde steps, such as **abandoning antiseptic treatment of wounds** and encouraging pus to form as part of the healing process.

Accurate marking of time

This reconstruction of de Dondi's 1364 astrarium shows three of its seven dials, as well as the balance wheel and weights that regulated its movement.



In 1364, the Italian clockmaker

Giovanni de Dondi (1318–89) published his

Planetarium, a description of the **complex astronomical clock (astrarium)** that he had just completed after

16 years of work. This 1 m- (3.3 ft-) high, weight-driven clock with an escapement and balance wheel was the most advanced of its time. Its **seven dials showed the celestial movements of the Sun, Moon, and five calendars**, and it acted as a perpetual calendar, including showing the date of Easter. The clock was regulated by a balance that swung 1,800 times an hour; and the addition or removal of small weights enabled corrections to be made if the device ran too fast or slow.

The **first recorded use of rockets** in Europe as a military weapon came in 1380 at the Battle of Chioggia, fought between the fleets of Venice and Genoa. Rockets need an ignition that provides continued and regular thrust as the projectile flies through the air (unlike cannon balls). Since gunpowder packed into a tube burns



Making gunpowder

German legend attributed the invention of gunpowder to alchemist Barthold Schwartz. This woodcut depicts him stirring together the ingredients for gunpowder.

unevenly and mostly at the surface, military technologists had to devise new techniques. They left a **conical hole in the centre of the tube**, which encouraged an even burn (and sufficient thrust), and **made the rocket airtight**, except for a small opening at the rear. These methods were discussed by the German military engineer **Konrad Kyeser** (1366–after 1405) in his *Bellifortis* (*War Fortifications*) in 1405. Kyeser also advised the adding of feathers (like the fletching of an arrow) or weights to the rear of the rocket to make its trajectory more even and to enable more accurate aiming.

1363 French physician Guy de Chauliac finishes his *Great Surgery*

1368 Fellowship of Surgeons recorded in London, England

1370 English surgeon John Arderne describes a new type of syringe

1377 Republic of Ragusa enacts quarantine laws

1380 First recorded use of rockets in Europe, at the Battle of Chioggia

1364 Italian clockmaker Giovanni de Dondi completes his astrarium clock after 16 years

1377 First effective use of cannons in European warfare at Oudenarde

1377 French philosopher Nicholas Oresme proposes idea that the Earth rotates on its axis

1383 Marseilles introduces quarantine regulations

1391 First dissection recorded in Spain